Everything You Need On a Single Board

This well-designed Z-80 computer from Colonial Data Systems Corp. is loaded with features and compares very favorably to other micros on the market.

By Terry Kepner

The SB-80 from Colonial Data Systems Corp. is one of the newest Z-80 computers on the market. It contains all the features you could want on a single board: 64K bytes of RAM memory, two parallel ports, two serial ports, single and double density floppy disk controller, clock circuit and power supply. Included in the computer's case are two Shugart eight-inch double-density disk drives.

One important aspect of the SB-80 is its 50-pin expansion connector which allows access to all data and address lines coming from the Z-80A. The obvious benefit of this is that it can easily support a host adapter interface for hard disks, expanded memory, additional serial and parallel ports, or any other hardware expansion. This gives it the capabilities of an S-100 bus machine while retaining the simplicity of a single-board computer.

The SB-80 is sold with the CP/M operating system for $2995. It can also operate with MP/M. It is a true CP/M computer and is capable of using almost any program based on CP/M.

Accounts Receivable, Accounts Payable, Mailing List, and many other business programs are available for the SB-80 from S & M Systems, 2 Washington St., Haverhill, MA 01830.

The SB-80 is sold without a terminal (keyboard and video) allowing you to buy only what you need for the system.

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System Features

The SB-80 (Colonial Data Systems Corp., 105 Sanford St., Hamden, CT 06514) is a single-board computer that uses the Z-80 CPU and operates at 4 MHz. When the system is first turned on, a built-in ROM performs a quick RAM check of the lower 16K bytes of RAM and then bootstraps the CP/M (or MP/M) disk into memory. After the ROM finishes the bootstrap, it "bank switches" itself out of the way leaving only RAM available.

RS-232. The RS-232 capability of the SB-80 is extensive. It has two completely independent, full duplex channels. Data rates can be selected from 50 to 19,200 baud. Each channel's receiver is quadruply buffered, with three eight-bit registers in a first-in-first-out arrangement and the fourth as an eight-bit input shift register. This quad buffering gives the CPU more time to service an interrupt before incoming serial data is lost because the CPU is busy elsewhere.

The transmitters are double buffered with one eight-bit output shift register and one eight-bit buffer register. Each channel has five eight-bit control registers; two eight-bit status registers and two eight-bit sync-character registers. There are also two 16-bit shift registers used for CRC (cyclical redundancy check) generation and checking with appropriate internal feedback. The feedback can be software controlled to use either one of two different CRC codes.

The RS-232 channels operate in any one of three different modes:

- Asynchronous with five, six, seven or eight bits/chr; one, ½, or two stop bits; even, odd or no parity; break generation and detection; parity, overrun and framing error detection; and ×1, ×16, ×32 and ×64 clock modes.
- Binary synchronous operation with internal or external character synchronization; one or two characters; sync characters in separate registers; automatic sync character insertion; and CRC generation and checking.
- HDLC (also called IBM SDLC) operation with automatic zero insertion and deletion; automatic flag insertion; address field recognition; I-field residue handling; valid receive messages protected from overrun; and CRC generation and checking.

In addition to these functions you also have daisychain-priority-interrupt logic so that you can use automatic interrupt vectoring without having to write your own external logic to handle this problem.

The physical setup of these ports is very practical. The RS-232 ports are connected to jumper pads on the circuit board, as are the connections to the DB-25 connectors at the rear of the computer's cabinet. The DB-25 pads and the RS-232 pads are connected together via jumper wires which you can easily change to any arrangement you need. This makes the task of connecting the SB-80 to non-standard RS-232 equipment simple; all you need to know are the pin connections of the hardware. How to do this is explained in the manual.

When the SB-80 is first powered up, channel A is set to 9600 baud and put in the asynchronous communication mode. Channel B is set to 300 baud.

There is one disadvantage to the setup of the RS-232 ports: there are only two pins available for the three signals T×CB, R×CB and DTRB. They are normally configured with T×CB and R×CB on the same pin because most communications will use the same clock speed and phase for both the transmit and receive modes. Because it's rare for the transmit and receive modes to operate at different speeds, I don't think that the lack of a separate pin for each of these three signals is much of a problem.

Parallel ports. There are two parallel ports in the SB-80; both are programmable and TTL compatible. The ports are eight-bit bidirectional, with handshake data control. The handshaking is driven by interrupts from the peripheral device connected to the computer.

There are four modes of operation:

- Byte output
- Byte input
- Bit control
- Byte bidirectional bus (port A only)

As with the serial ports, the parallel ports incorporate daisychain-priority-interrupt logic for automatic interrupt vectoring.

On the parallel ports this interrupt priority means that you can have two terminals connected simultaneously with the computer servicing each terminal only when it receives an interrupt from that terminal. This prevents it from spending half its time checking each terminal to see if there is anything waiting for it.

The port logic is divided into four sections: the control logic, the interrupt control logic, port A I/O logic and port B I/O logic. Each port's logic control is composed of six registers: an eight-bit data-input register, an eight-bit output register, a two-bit mode control register, an eight-bit mask register and eight-bit input/output select register and a two-bit mask control register.

Colonial Data's single-board Z-80 computer, the SB-80.

The mode control register is used to select one of the four possible operation modes (byte input, byte output, byte bidirectional or bit control). Data transfer between the CPU and the peripheral is through the I/O registers. The eight-bit mask register, the eight-bit I/O select registers and the two-bit mask control register are reserved for use only in the bit-control mode. As in a security system with triggers which are either on or off, the bit mode is a very powerful method of connecting the SB-80 to other devices which generate only one condition.

The I/O register allows you to specify each individual bit in the mask register as either an input or an output bit. The mask register itself determines which bits are to be scanned. The two-bit control register lets you specify if the incoming/outgoing bit is going to be either a logical 1 or a logical 0. Therefore, if you have the SB-80 connected to an alarm system of five alarms, the SB-80 doesn't have to poll the alarms one at a time. It works on other tasks instead and only looks at the alarms when it receives an interrupt signal that tells it to scan the parallel port to see which alarm (or any number of them) has sent in a signal of activity.

Timer circuit. The on-board counter/timer is a programmable, four-channel device that provides the counting and timing functions of the SB-80. Each channel is composed of two registers, two counters and their control logic. The two registers are an eight-bit, time-constant register which initializes and re-loads the down-counter at the start. Each time the down-counter reaches 0 there is an eight-bit control register which selects the mode and conditions of the channel's operation. The counters are an eight-bit down-counter which takes the number given to it by the time-constant register and decreases it until it reaches 0, and an eight-bit pre-scaler that can be programmed to divide the 4 MHz clock rate of the computer by either 16 or 256. The pre-scaler determines the rate at which the down-counter decreases.

Expansion adapter. The 50-pin expansion adapter makes the Z-80 CPU bus available for further growth of the SB-80. It has a few nice features in addition to the buffering and availability of the Z-80 Tri-state bus. One of these features is an 8 MHz clock at the connector. This extra clock operates at twice the rate of the system clock and allows you more complex clocking circuitry in external devices than would otherwise be possible.

Another feature allows external circuitry to take control of the internal Z-80 bus if you need it. Another available feature allows you to lock out the internal memory of the SB-80 and lets the external circuitry supply the memory to be used by the SB-80. This provides a simple and effective method of "bank switching" memory into the SB-80 at the control of external circuitry.

Without having to create external

logic, you can daisychain up to four Z-80 peripheral chips into the priority interrupt structure. On-board logic assures that the highest priority device which requests an interrupt will be serviced first. If you find it necessary to add more than four Z-80 chips to the expansion adapter, you can add your own "look ahead" logic and connect up to 30 chips using standard TTL logic circuits.

The SB-80 isn't directly S-100 compatible because all the control and data lines are brought out on the expansion bus and properly buffered. However, it's an easy chore to interface the SB-80 to almost any device on the market. Although it isn't mentioned in the documentation or advertisements, one of the immediate uses of the expansion adapter is the ability to interface the SB-80 with Corvus and Shugart hard disks giving you an option to add up to 80 megabytes of hard-disk storage capacity. In fact, according to Bob Schock, the president of Colonial Data, many of their computers are sold with hard-disk drives as a package deal.

Keyboard interface. The keyboard interface is a simple 8 × 8 switch matrix capable of being connected to a stand-alone keyboard of 62 keys. The

SYSTEM	TYPE	CPU	BITS	SPEED	OPSYS	LANGUAGE	TIME	TESTED BY:
COLONIAL DATA	SB80	Z80	8	4	CP/M2.2	MBasic5.2	661	S & M Systems
ALTOS	8002	Z80	8	4	CP/M2.2	MBasic5.2	662	S & M Systems
CA. COMPSYS	2810	Z80	8	4	CP/M2.2	MBasic5.2	663	Bob Loesch
OHIO SCI	C4-P	6502	8	2	OS65D3.2	Level I Basic	680	
CROMENCO	Z-2H	Z80	8	4	CDOS2.36	32K Basic, SFP	935	Paul Hansknecht
TANDY	TRS-80II	Z80	8	4	TRSDOS	Disk Basic	955	
APPLE	IIPLUS	6502	8	2	DOS 3.2	APLSOFTIIBasic	960	
CROMENCO	Z-2H	Z80	8	4	CDOS2.36	32K Basic,LFP	1130	Paul Hansknecht
OHIO SCI	C3-C	6502	8	1	OS65D	Level I Basic	1346	
HP	HP-85	PROP	8	N/A	N/A	Basic	1380	
BASIC/FOUR	600	8080	8	N/A	N/A	Basic	1404	
ZENITH	Z-89	Z80	8	2	CP/M2.2	MBasic5.2	1500	S & M Systems
IMSAI	I8080	8080	8	2	CP/M2.2	MBasic50	1614	
TANDY	TRS-80III	Z80	8	2	TRSDOS	Disk Basic	1695	William Gollan
EXIDY	SORC'R	Z80	8	2	CP/M1.4	MBasic50	1740	Henry Deutsch
TANDY	TRS-80I	Z80	8	2	TRSDOS	LevelIIBasic	1928	

Table 1. Benchmark test results.

software, using this technique, has complete control of the keyboard including the ability to detect the depression of more than one key.

Floppy disk controller. The SB-80 uses the Western Digital FD1793 for dual-density data bus control and is compatible with the IBM 3740 and the IBM System 34.

The manual. The documentation supplied with the SB-80 is in two, two-inch thick, three-ring binders. The first binder covers the hardware and initial operation of the SB-80; the second binder covers the CP/M operating system that comes with the unit. Binder number one is roughly divided into three sections. The first section covers the main hardware features of the SB-80: first in a brief form that summarizes each feature's attributes, and then in more specific detail.

The first section is about 68 pages and includes seven fold-out pages of hardware schematics. The second section, 25 pages, is the operator's guide to using the SB-80. It tells you how to power it up, what the connectors on the back do and the functions of the lights, switches and doors on the front of the unit. It lists the procedures necessary to turn on the SB-80 and how to make backups of your CP/M system disks. It also gives a brief overview of the CP/M system. The third section is a technical section which includes schematics covering the Shugart disk drives and Boschert power supply sold with the system.

The faults I find with the manual are the same as I find with most manuals. They seem to be written by technical people who assume that you already are familiar with their system, or that you have experience with computers of this type. Technicians will find the manual to be very comprehensive. It supplies all the information necessary to get the SB-80 to do what they want it to do and keep it operational. It is not written for the computer novice.

Colonial Data Systems includes a brief tutorial (25 pages) on the CP/M operating system. They also include an on-line CP/M help command. Typing the word help followed by the command you want explained will result in a display of the definition of the command. They also include a program on their distribution disk called "Help Help" which gives descriptions of the programs on the disk.

To show you how the SB-80 computer compares to the other computers on the market, I included a benchmark test (shown in Table 1, courtesy of Bill Gollan and S & M Systems). All benchmarks were run with a Basic interpreter. Compilers were not used because of their obvious speed advantage, and because interpreters are more available and represent a much wider group of machines. The CPUs of the computers are listed along with the clock speeds (in MHz) at which the computers operate. The operating systems and versions of Basic used are also listed. The last item of each line is the name of the person who actually tested the computer.

The benchmark test is a simple one and uses two nested FOR-NEXT loops and IF-THEN tests. The program used was:

```
140 FOR N=1 TO 1000
150 FOR K=2 to 500
160 LET M=N/K
170 LET L=INT(M)
180 IF L=0 THEN 230
190 IF L=1 THEN 220
200 IF M>L THEN 220
210 IF M=L THEN 240
220 NEXT K
230 PRINT N;
240 NEXT N
250 PRINT "FINISHED"
```

The results of the benchmark test are listed in seconds.

Summary

The SB-80 computer is a well-planned and designed Z-80 computer. Its designers have obviously spent a lot of time and effort in laying out the SB-80 for maximum versatility and convenience. The design of the pin-to-pad RS-232 lines is a much needed setup. More than once I have wanted to connect a standard RS-232 printer to a standard RS-232 port but ran into incompatibility problems. The design of the SB-80 RS-232 ports alleviates this problem.

Another advantage of the SB-80 is the multitude of ready-to-use programs for the CP/M operating system. Although the SB-80 is a new machine, it is not limited by a lack of software.

As you can tell, I am enthusiastic about the capabilities of the SB-80 computer. I think it is definitely one of the better, if not the best, Z-80 computers to be released in the last few years. ■